P05. Hybrid Reinforcement Learning–Based Eco-Driving Strategy for Connected and Automated Vehicles at Signalized Intersections

Zhengwei Bai; Peng Hao; Wei Shangguan; Baigen Cai; Matthew J. Barth   2022

IEEE Transactions on Intelligent Transportation Systems   PDF pages: 15   Relevance: 86/100

1. 论文全景速览与核心贡献 / High-Level Overview & Core Contributions

研究背景与痛点 / Background & Pain Points

这篇论文位于“Eco-driving + hybrid RL”方向。对你的 JSSP-based Intersection Management 课题，它回答的是高层调度、低层轨迹平滑、安全过滤或真实验证中的一个关键子问题：如何让车辆在路口少停、少冲突、少能耗，同时保持在线可执行。

The paper belongs to the theme of Eco-driving + hybrid RL. For a JSSP-based intersection-management thesis, it illuminates one of the key layers: scheduling, smoothing, safety filtering, or real-world validation.

Signalized setting and policy-level action output differ from a scheduler-to-smoother architecture.

核心科学问题 / Core Scientific Question

在信号灯、V2I 信息和混合交通干扰下，如何通过混合规则-学习策略降低能耗和时间损失?

How can a hybrid rule-learning eco-driving policy reduce energy and travel time under SPaT and mixed-traffic constraints?

科学贡献 / Scientific Contributions

- 方法贡献 (method contribution)：Combines a rule-based driving manager, V2I/vision features, and RL to generate longitudinal and lateral eco-driving actions.
- 安全/避碰贡献 (safety contribution)：Mixed-traffic policy includes rule-based safety management around surrounding vehicles.
- 平滑/停止避免贡献 (smoothing contribution)：Targets energy and travel-time reduction by smoothing approach behavior at signalized intersections.
- 创新力度评判 (reviewer judgement)：强应用价值：规则管理器增强 RL 安全性，适合作为轨迹平滑器能耗目标的参考。

核心概念对照 / Core Concepts

中文概念	English Concept	Role / 学术作用
自动交叉口管理	Autonomous Intersection Management	把信号控制替换为车辆级调度与控制。
轨迹平滑	Trajectory Smoothing	减少急加减速、停车和能耗峰值。
安全约束	Safety Constraints	把碰撞风险写成距离、时间间隔或屏障函数。
Hybrid RL eco-driving at signalized intersections	信号交叉口混合强化学习生态驾驶	该论文的核心方法族。

教材式学习路径 / Textbook-Style Study Path

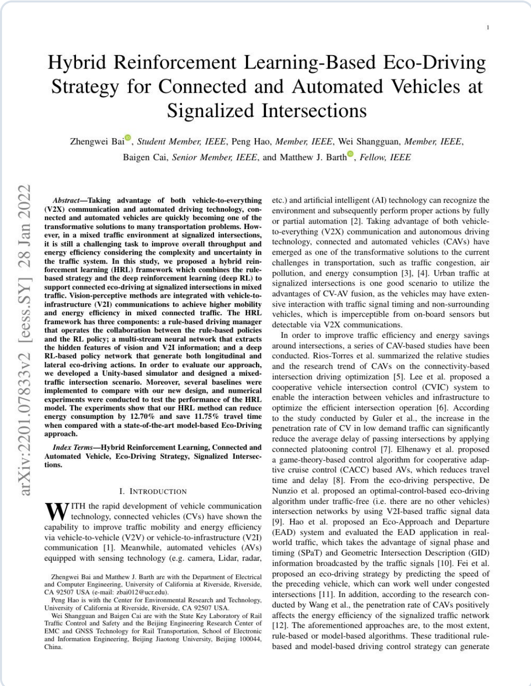
- 第一遍先读首页截图、摘要与引言，确认研究问题和场景假设。
- 第二遍沿着技术路线图复现模型变量、约束和求解器输入输出。
- 第三遍逐节阅读 section deep reading，对照 PDF 截图检查公式、图表和实验。
- 第四遍只站在审稿人视角重读实验，判断 baseline、公平性、消融和鲁棒性。
- 最后把博士 checklist 写成自己的复现实验或论文拓展计划。

博士阅读 Checklist / Doctoral Reading Checklist

- 能否独立重写核心公式：能耗-效率-安全复合奖励 / Energy-efficiency-safety reward，并解释每个符号的物理意义?
- 能否画出输入-模型-算法-控制-验证的数据流图?
- 能否指出至少两个强假设，并说明它们在真实交通中何时失效?
- 能否复述实验基准是否公平，指标是否覆盖效率、安全、能耗和计算时间?
- 能否提出一个与自己 JSSP/RL/smoother 课题直接相连的拓展实验?

关键截图讲解 / PDF Snapshot Explanation

截图来自本地 PDF 的精选页面，用于学习定位和版面理解；请与原 PDF 平行阅读，不把截图视作全文替代。



Hybrid Reinforcement Learning-Based Eco-Driving Strategy for Connected and Automated Vehicles at Signalized Intersections

Zhengwei Bai, *Student Member, IEEE*, Peng Hao, *Member, IEEE*, Wei Shangguan, *Member, IEEE*, Baigen Cai, *Senior Member, IEEE*, and Matthew J. Barth, *Fellow, IEEE*

arXiv:2201.07833v2 [eess.SY] 28 Jan 2022

I. Introduction

首页与摘要 / Title and abstract page

p.1 · full_page

这张截图用于把笔记中的抽象概念拉回原论文页面，便于你平行阅读原文、公式、图表和实验设置。

This snapshot anchors the study note to the original PDF page so you can read the explanation alongside the source layout.

与当前课题的连接：Supports the argument that stop-and-go reduction and energy-aware smoothing are important low-level objectives.

II. Problem Formulation

建模或问题设定页 / Modeling or problem-setup page

p.2 · full_page

这张截图用于把笔记中的抽象概念拉回原论文页面，便于你平行阅读原文、公式、图表和实验设置。

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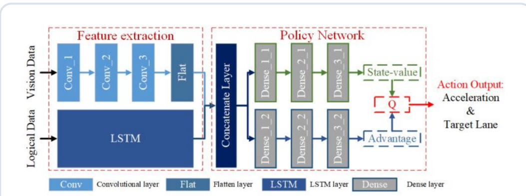

Fig. 7: The architecture of the neural network.

实验或结果页 / Experiments or results page

p.9 · figure_crop

这张截图用于把笔记中的抽象概念拉回原论文页面，便于你平行阅读原文、公式、图表和实验设置。

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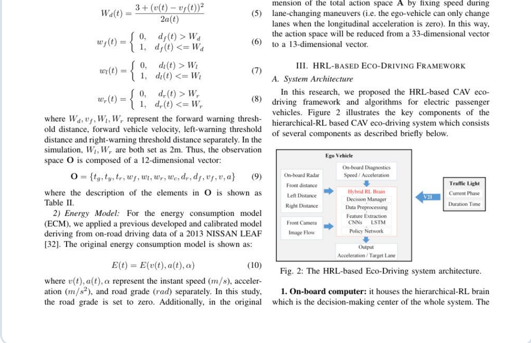

$$W_d(t) = \frac{3+(v(t)-v_f(t))^2}{2a(t)} \quad (5)$$

$$w_f(t) = \begin{cases} 0, & d_f(t) > W_d \\ 1, & d_f(t) <= W_d \end{cases} \quad (6)$$

$$w_l(t) = \begin{cases} 0, & d_l(t) > W_l \\ 1, & d_l(t) <= W_l \end{cases} \quad (7)$$

$$w_r(t) = \begin{cases} 0, & d_r(t) > W_r \\ 1, & d_r(t) <= W_r \end{cases} \quad (8)$$

where W_d, v_f, W_l, W_r represent the forward warning threshold distance, forward vehicle velocity, left-warning threshold distance and right-warning threshold distance separately. In the simulation, W_l, W_r are both set as 2m. Thus, the observation space $\mathbf{O}$ is composed of a 12-dimensional vector:

$$\mathbf{O} = \{t_g, t_y, t_r, w_f, w_l, w_r, w_c, d_r, d_f, v_f, v, a\} \quad (9)$$

where the description of the elements in $\mathbf{O}$ is shown as Table II.

2) *Energy Model:* For the energy consumption model (ECM), we applied a previous developed and calibrated model deriving from on-road driving data of a 2013 NISSAN LEAF [32]. The original energy consumption model is shown as:

$$E(t) = E(v(t), a(t), \alpha) \quad (10)$$

where $v(t), a(t), \alpha$ represent the instant speed (m/s), acceleration (m/s^2), and road grade (rad) separately. In this study, the road grade is set to zero. Additionally, in the original

mension of the total action space $\mathbf{A}$ by fixing speed during lane-changing maneuvers (i.e. the ego-vehicle can only change lanes when the longitudinal acceleration is zero). In this way, the action space will be reduced from a 33-dimensional vector to a 13-dimensional vector.

III. HRL-BASED ECO-DRIVING FRAMEWORK

A. System Architecture

In this research, we proposed the HRL-based CAV eco-driving framework and algorithms for electric passenger vehicles. Figure 2 illustrates the key components of the hierarchical-RL based CAV eco-driving system which consists of several components as described briefly below.

Fig. 2: The HRL-based Eco-Driving system architecture.

1. On-board computer: it houses the hierarchical-RL brain which is the decision-making center of the whole system. The

核心方法或算法页 / Core method or algorithm page

p.4 · full_page

这张截图用于把笔记中的抽象概念拉回原论文页面，便于你平行阅读原文、公式、图表和实验设置。

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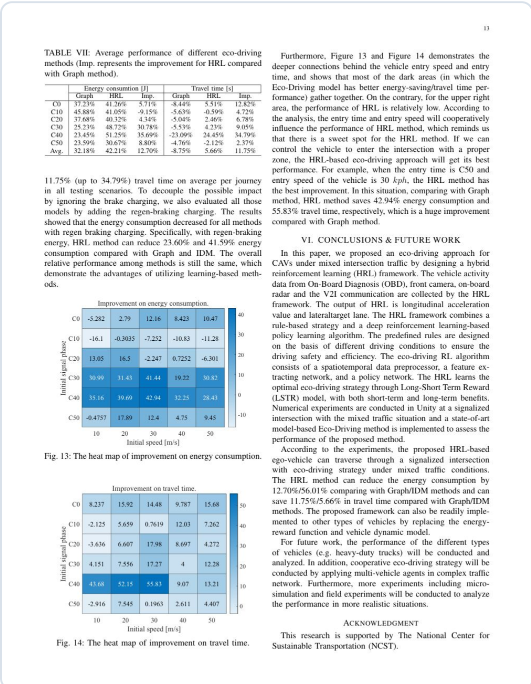

TABLE VII: Average performance of different eco-driving methods (Imp. represents the improvement for HRL compared with Graph method).

	Energy consumtion [J]			Travel time [s]		
	Graph	HRL	Imp.	Graph	HRL	Imp.
C0	37.23%	41.26%	5.71%	-8.44%	5.51%	12.82%
C10	45.88%	41.05%	-9.15%	-5.63%	-0.59%	4.72%
C20	37.68%	40.32%	4.34%	-5.04%	2.46%	6.78%
C30	25.23%	48.72%	30.78%	-5.53%	4.23%	9.05%
C40	23.45%	51.25%	35.69%	-23.09%	24.45%	34.79%
C50	23.59%	30.67%	8.80%	-4.76%	-2.12%	2.37%
Avg.	32.18%	42.21%	12.70%	-8.75%	5.66%	11.75%

11.75% (up to 34.79%) travel time on average per journey in all testing scenarios. To decouple the possible impact by ignoring the brake charging, we also evaluated all those models by adding the regen-braking charging. The results showed that the energy consumption decreased for all methods with regen braking charging. Specifically, with regen-braking energy, HRL method can reduce 23.60% and 41.59% energy consumption compared with Graph and IDM. The overall relative performance among methods is still the same, which demonstrate the advantages of utilizing learning-based methods.

Fig. 13: The heat map of improvement on energy consumption.

Fig. 14: The heat map of improvement on travel time.

Furthermore, Figure 13 and Figure 14 demonstrates the deeper connections behind the vehicle entry speed and entry time, and shows that most of the dark areas (in which the Eco-Driving model has better energy-saving/travel time performance) gather together. On the contrary, for the upper right area, the performance of HRL is relatively low. According to the analysis, the entry time and entry speed will cooperatively influence the performance of HRL method, which reminds us that there is a sweet spot for the HRL method. If we can control the vehicle to enter the intersection with a proper zone, the HRL-based eco-driving approach will get its best performance. For example, when the entry time is C50 and entry speed of the vehicle is 30 kph, the HRL method has the best improvement. In this situation, comparing with Graph method, HRL method saves 42.94% energy consumption and 55.83% travel time, respectively, which is a huge improvement compared with Graph method.

VI. CONCLUSIONS & FUTURE WORK

In this paper, we proposed an eco-driving approach for CAVs under mixed intersection traffic by designing a hybrid reinforcement learning (HRL) framework. The vehicle activity data from On-Board Diagnosis (OBD), front camera, on-board radar and the V2I communication are collected by the HRL framework. The output of HRL is longitudinal acceleration value and lateraltarget lane. The HRL framework combines a rule-based strategy and a deep reinforcement learning-based policy learning algorithm. The predefined rules are designed on the basis of different driving conditions to ensure the driving safety and efficiency. The eco-driving RL algorithm consists of a spatiotemporal data preprocessor, a feature extracting network, and a policy network. The HRL learns the optimal eco-driving strategy through Long-Short Term Reward (LSTR) model, with both short-term and long-term benefits. Numerical experiments are conducted in Unity at a signalized intersection with the mixed traffic situation and a state-of-art model-based Eco-Driving method is implemented to assess the performance of the proposed method.

According to the experiments, the proposed HRL-based ego-vehicle can traverse through a signalized intersection with eco-driving strategy under mixed traffic conditions. The HRL method can reduce the energy consumption by 12.70%/56.01% comparing with Graph/IDM methods and can save 11.75%/5.66% in travel time compared with Graph/IDM methods. The proposed framework can also be readily implemented to other types of vehicles by replacing the energy-reward function and vehicle dynamic model.

For future work, the performance of the different types of vehicles (e.g. heavy-duty trucks) will be conducted and analyzed. In addition, cooperative eco-driving strategy will be conducted by applying multi-vehicle agents in complex traffic network. Furthermore, more experiments including micro-simulation and field experiments will be conducted to analyze the performance in more realistic situations.

ACKNOWLEDGMENT

This research is supported by The National Center for Sustainable Transportation (NCST).

结论或局限页 / Conclusion or limitations page

p.13 · full_page

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章节精读 / Section-by-Section Deep Reading

p.1 · motivation

I. INTRODUCTION

章节目标： 建立研究动机：为什么传统信号控制、FCFS 或单车控制不足以处理该论文关注的交叉口协同问题。

Learning goal: Understand why the paper needs a new coordination method rather than a standard signal, FCFS, or single-vehicle controller.

关键论点 / Key argument: 这一部分通常把痛点落到 Eco-driving + hybrid RL：既要提高效率，又要保持安全与在线可执行。

技术焦点 / Technical focus: 精读时标出作者批判的 baseline、场景假设、交通参与者类型和隐含优化目标。

审稿追问 / Reviewer question: 作者指出的痛点是否真的需要新方法，还是可以由更强的优化器/更保守规则解决?

p.2 · model

II. PROBLEM FORMULATION

章节目标： 把自然语言交通问题压缩为变量、状态、约束、目标函数或图结构。

Learning goal: Translate the traffic problem into variables, states, constraints, objectives, or graph relations.

关键论点 / Key argument: 本节是复现论文的核心入口，应与公式“能耗-效率-安全复合奖励 / Energy-efficiency-safety reward”一起读。

技术焦点 / Technical focus: 逐项检查车辆状态、冲突关系、控制输入、时间/空间索引和安全阈值是否定义完整。

审稿追问 / Reviewer question: 这些建模假设在混合交通、通信延迟、感知误差和高密度车流下是否仍成立?

p.4 · method

III. HRL- BASED ECO-D RIVING FRAMEWORK

章节目标： 解释算法如何把模型转化为可执行决策，并说明在线运行机制。

Learning goal: Trace how the paper turns the model into executable online decisions.

关键论点 / Key argument: 系统用规则型驾驶管理器处理安全和交通规则，用强化学习在允许动作空间中选择纵向/横向生态驾驶动作；V2I 和视觉特征帮助策略预见信号相位与周围车辆。

技术焦点 / Technical focus: 画出输入、核心求解器、输出和安全检查接口，明确哪些步骤是优化、哪些是规则、哪些是学习。

审稿追问 / Reviewer question: 若算法某一步失败，论文有没有 fallback、重规划或安全过滤机制?

p.4 · bridge

1. On-board computer: it houses the hierarchical-RL brain

章节目标： 承接前后章节，补充局部定义、案例或实现细节。

Learning goal: Recover implementation details needed for reproduction.

关键论点 / Key argument: 这类章节常常不是主贡献，但决定你能否真正复现方法。

技术焦点 / Technical focus: 检查它是否给出了参数、伪代码、场景划分或实现细节。

审稿追问 / Reviewer question: 跳过该节会不会导致公式或实验无法复现?

p.5 · bridge

p.5 · bridge

2. On–board radar : As defined in previous section, three

章节目标：承接前后章节，补充局部定义、案例或实现细节。

Learning goal: Recover implementation details needed for reproduction.

关键论点 / Key argument: 这类章节常常不是主贡献，但决定你能否真正复现方法。

技术焦点 / Technical focus: 检查它是否给出了参数、伪代码、场景划分或实现细节。

审稿追问 / Reviewer question: 跳过该节会不会导致公式或实验无法复现?

3. On–board camera : this is installed in the front of

章节目标：承接前后章节，补充局部定义、案例或实现细节。

Learning goal: Recover implementation details needed for reproduction.

关键论点 / Key argument: 这类章节常常不是主贡献，但决定你能否真正复现方法。

技术焦点 / Technical focus: 检查它是否给出了参数、伪代码、场景划分或实现细节。

审稿追问 / Reviewer question: 跳过该节会不会导致公式或实验无法复现?

p.5 · bridge

4. On–board diagnostics (OBD) : this component can get

章节目标：承接前后章节，补充局部定义、案例或实现细节。

Learning goal: Recover implementation details needed for reproduction.

关键论点 / Key argument: 这类章节常常不是主贡献，但决定你能否真正复现方法。

技术焦点 / Technical focus: 检查它是否给出了参数、伪代码、场景划分或实现细节。

审稿追问 / Reviewer question: 跳过该节会不会导致公式或实验无法复现?

p.9 · experiment

IV. G AME ENGINE BASED SIMULATION EXPERIMENTS

章节目标：验证方法是否在公平基准下改善效率、安全、能耗或计算时间。

Learning goal: Check whether the claimed improvements are supported by fair baselines and meaningful metrics.

关键论点 / Key argument: 实验应与论文声称的贡献闭环：Unity–based mixed–traffic intersection simulation; reports 12.70 percent energy reduction and 11.75 percent travel–time saving against a model–based eco–driving baseline.

技术焦点 / Technical focus: 重点追踪指标：能耗降低百分比、行程时间、停车次数、舒适性指标和碰撞/违规事件。 同时检查仿真平台、交通需求和随机性设置。

审稿追问 / Reviewer question: 实验是否只展示平均收益，还是覆盖了高密度、异常扰动和失败案例?

逐段式导读 / Paragraph-Level Walkthrough

Opening motivation / 开篇动机段

先把作者的痛点翻译成一句博士问题：在信号灯、V2I 信息和混合交通干扰下，如何通过混合规则-学习策略降低能耗和时间损失？ 读这一段时不要急着接受动机，要追问传统 FCFS、信号灯或保守 MPC 为什么不足。

Turn the opening motivation into one doctoral-level question and test whether the paper truly needs a new method.

Assumption and setting / 假设与场景段

把车辆类型、通信条件、路口类型和动力学假设列成表。该文的关键边界包括：可获得信号相位与配时 (SPaT) 或可靠 V2I 信息。；规则管理器覆盖主要危险场景。；能耗模型与仿真平台对真实车辆足够近似。

List vehicle type, communication, intersection setting, and dynamics assumptions before reading the equations.

Model and notation / 模型与符号段

围绕核心公式“能耗-效率-安全复合奖励 / Energy-efficiency-safety reward”复写符号表，确认每个变量是决策变量、状态变量、参数还是约束阈值。

Rewrite the symbol table around the core formulation and classify variables as states, decisions, parameters, or constraints.

Algorithm mechanism / 算法机制段

把算法读成一条数据流：输入交通状态，执行 Combines a rule-based driving manager, V2I/vision features, and RL to generate longitudinal and lateral eco-driving actions.，输出可执行通行/控制决策，并检查安全接口。

Read the algorithm as a dataflow from traffic state to executable sequence/control decision, with explicit safety interfaces.

Experiment and evidence / 实验证据段

不要只看作者报告的提升，要检查基准、公平性和指标。本文验证描述为：Unity-based mixed-traffic intersection simulation; reports 12.70 percent energy reduction and 11.75 percent travel-time saving against a model-based eco-driving baseline.

Go beyond reported gains and inspect baselines, fairness, metrics, and computation time.

Reviewer synthesis / 审稿式总结段

最后把贡献拆成可发表与需补强两部分。对你课题最直接的用途是：Supports the argument that stop-and-go reduction and energy-aware smoothing are important low-level objectives.

Separate publishable contribution from missing evidence, then connect the result to your own thesis architecture.

技术路线图与贡献量评估 / Technical Route & Contribution Matrix

1. 输入
Input

车辆状态、路径/车道、交通需求、信号或协同信息。

2. 建模
Modeling

能耗–效率–安全复合奖励 / Energy–efficiency–safety reward

3. 核心求解
Core solver

Combines a rule–based driving manager, V2I/vision features, and RL to generate longitudinal and lateral eco–driving actions.

4. 安全接口
Safety interface

Mixed–traffic policy includes rule–based safety management around surrounding vehicles.

5. 平滑与效率
Smoothing and efficiency

Targets energy and travel–time reduction by smoothing approach behavior at signalized intersections.

6. 验证
Validation

Unity–based mixed–traffic intersection simulation; reports 12.70 percent energy reduction and 11.75 percent travel–time saving against a model–based eco–driving baseline.

贡献量评估 / Contribution Strength Matrix

Dimension / 维度	Score / 评分	Comment / 评语
理论贡献 / Theoretical contribution	3/5	能耗–效率–安全复合奖励 / Energy–efficiency–safety reward
算法贡献 / Algorithmic contribution	5/5	Combines a rule–based driving manager, V2I/vision features, and RL to generate longitudinal and lateral eco–driving actions.
实验贡献 / Experimental contribution	3/5	Unity–based mixed–traffic intersection simulation; reports 12.70 percent energy reduction and 11.75 percent travel–time saving against a model–based eco–driving baseline.
工程贡献 / Engineering contribution	4/5	Mixed–traffic policy includes rule–based safety management around surrounding vehicles.
博士课题价值 / PhD relevance	4/5	Supports the argument that stop–and–go reduction and energy–aware smoothing are important low–level objectives.

2. 理论方法论深度解构 / Deep Dive into Methodology & Theoretical Framework

问题数学建模 / Mathematical Formulation

能耗-效率-安全复合奖励 / Energy-efficiency-safety reward

$$
\max_{\pi}\; \mathbb{E}\sum_t \gamma^t r_t,\quad r_t= -w_e E_t-w_d D_t-w_s \mathbb{I}_{unsafe}-w_a |a_t|
$$

Symbol / 符号	含义	Meaning	Role / 建模作用
E_t	瞬时能耗	instantaneous energy use	生态驾驶优化主目标
D_t	延误/时间损失	delay or time loss	避免单纯省能导致低速拖行
I_{unsafe}	不安全事件指示	unsafe-event indicator	对碰撞或近碰撞施加强惩罚
a_t	加速度动作	acceleration action	连接舒适性和平滑性
w_e,w_d,w_s,w_a	奖励权重	reward weights	决定策略偏好

核心算法架构 / Core Algorithm Layout

系统用规则型驾驶管理器处理安全和交通规则，用强化学习在允许动作空间中选择纵向/横向生态驾驶动作；V2I 和视觉特征帮助策略预见信号相位与周围车辆。

A rule-based driving manager constrains behavior, while RL chooses eco-driving actions using SPaT/V2I and perception features.

> **收敛性、稳定性或实时性 / Convergence, Stability, Real-Time Feasibility**
>
> 规则层提高可控性和可解释性，但策略性能仍依赖训练分布与奖励权重。相比纯 RL，它更适合工程系统中的安全外壳。
>
> The rule layer improves interpretability and reduces unsafe exploration, making the hybrid policy more deployable than a purely learned controller.

潜在假设与边界条件 / Assumptions & Boundary Conditions

- 可获得信号相位与配时 (SPaT) 或可靠 V2I 信息。
- 规则管理器覆盖主要危险场景。
- 能耗模型与仿真平台对真实车辆足够近似。

3. 实验验证与结果评判 / Experimental Validation & Result Evaluation

基准对比与环境 / Baselines & Environment

模型式生态驾驶、普通 RL、人工驾驶/IDM、固定速度或信号响应策略。

Unity-based mixed-traffic intersection simulation; reports 12.70 percent energy reduction and 11.75 percent travel-time saving against a model-based eco-driving baseline.

核心指标 / Metrics

能耗降低百分比、行程时间、停车次数、舒适性指标和碰撞/违规事件。

消融实验 / Ablation

需要分别去掉规则管理器、V2I 特征、视觉特征和横向动作，才能说明混合结构的真实贡献。

博士阅读提醒 / Reading Reminder

阅读实验时不要只看平均值；应检查交通密度、车流方向、随机种子、失败案例和计算耗时。For doctoral reading, inspect density regimes, demand patterns, random seeds, failure cases, and computation time rather than only mean improvements.

图表阅读线索 / Figure and Table Reading Cues

PDF extraction detected approximately 40 figure references and 8 table references. Exact captions remain in the original PDF; the bilingual cues below tell you what to inspect first.

图表名称	Figure/Table Name	Reading focus / 阅读重点
系统框架图	system framework diagram	先确认输入、决策层、控制层和安全约束之间的数据流。
策略网络/训练流程图	policy-network or training-pipeline diagram	检查状态、动作、奖励、经验回放和安全惩罚是否闭环一致。
实验指标对比图表	experimental metric comparison plots/tables	重点看平均值之外的密度变化、失败场景、计算时间和方差。

4. 审稿人视角：批判性思考与未来方向 / Critical Thinking & Future Directions

论文局限性 / Limitations & Weaknesses

- 信号化场景与无信号 AIM 调度问题不同。
- Unity 仿真能否代表真实车辆能耗仍需校准。
- 奖励权重敏感，跨路口迁移可能不稳定。

博士课题拓展点 / Research Extensions for PhD

- 把能耗奖励移植到 JSSP smoother，作为低层轨迹成本。
- 用多目标 RL 生成 Pareto 前沿，而不是固定奖励权重。
- 在 SUMO/真实轨迹数据上校准能耗模型，提高可发表可信度。

如何服务当前课题 / Use in This Project

Supports the argument that stop-and-go reduction and energy-aware smoothing are important low-level objectives.

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5. 交互式可视化与 PDF 排版蓝图 / Interactive Visualization & PDF Layout Blueprint

动态参数探索 / Parameter Exploration

能耗权重 / Energy weight:

1.0

能耗权重越高，轨迹越平滑，但可能牺牲通行时间。

Higher energy weight smooths trajectories but can increase travel time.

HTML 结构化布局建议 / HTML Structuring

- 顶部固定 metadata bar：题名、作者、年份、主题、PDF 页数。
- 左侧目录/section navigator，右侧为五大精读模块。
- 公式卡片使用 <pre> 或 LaTeX block，紧跟符号表。
- 审稿人批注用 reviewer-note block，博士拓展用 research-idea list。
- 交互参数用 input[type=range] + canvas 曲线，打印 PDF 时隐藏交互控件但保留解释。

来源锚点与逐节阅读路径 / Source Anchors & Section-by-Section Reading Map

以下锚点来自本地 PDF 抽取，用于回到原文复核；笔记不保存论文全文。

Page	Extracted heading
p.1	I. INTRODUCTION
p.2	II. PROBLEM FORMULATION
p.9	IV. G AME ENGINE BASED SIMULATION EXPERIMENTS

p.1 · I. INTRODUCTION

定位问题背景、研究缺口和为什么现有保守/非协同方法不足。

Frames the motivation, research gap, and limits of conservative or non-cooperative methods.

p.2 · II. PROBLEM FORMULATION

把交通交互转写为状态、约束、目标或图结构，是精读时最应复现的部分。

Defines states, constraints, objectives, or graph structures; this is the section to reproduce carefully.

p.4 · III. HRL- BASED ECO-D RIVING FRAMEWORK

给出算法流程和求解机制，应重点追踪输入、输出和安全接口。

Gives the algorithmic mechanism; track inputs, outputs, and safety interfaces.

p.4 · 1. On-board computer: it houses the hierarchical-RL brain

作为局部论证段落阅读，关注它如何服务主问题。

Read as a local argument and ask how it supports the main question.

p.5 · 2. On-board radar : As defined in previous section, three

作为局部论证段落阅读，关注它如何服务主问题。

Read as a local argument and ask how it supports the main question.

p.5 · 3. On-board camera : this is installed in the front of

作为局部论证段落阅读，关注它如何服务主问题。

Read as a local argument and ask how it supports the main question.

p.5 · 4. On-board diagnostics (OBD) : this component can get

作为局部论证段落阅读，关注它如何服务主问题。

Read as a local argument and ask how it supports the main question.

p.9 · IV. G AME ENGINE BASED SIMULATION EXPERIMENTS

检验方法是否真的改善效率、安全或能耗；要关注基准是否公平。

Tests efficiency, safety, or energy claims; inspect whether baselines are fair.